

Lecture 9

Compactness

- $A \subset M$ metric space
- 2 definitions
 - A is compact if every sequence in A has a convergent subsequence with its limit in A
 - A is covering compact if every covering of A reduces to a finite covering of A

Theorem 1: The two definitions above are equivalent

← direction

- (a_n) is a sequence in A
- Then, $\nexists$ subsequence converging in A
- WLOG, a_n does not have repetition
- $\exists M_{r(p)}(p)$ containing only finitely many a_n
 - If it has infinitely many a_n , we have convergence
- $\{M_r(p)\} =$ the open covering of A
- Reduce to a finite subcovering
- Each neighborhood has finitely many terms of the sequence, so there are a finite number of terms in total
- $\Downarrow$ to (a_n) is a sequence

→ direction

- Needs a lemma, specifically Lebesgue numbers
- Lebesgue number of a covering $\mathcal{U}$ of A (but also a collection of open sets) is a λ s.t. $\forall p \in A$, $M_\lambda(p) \subset \text{some } U \in \mathcal{U}$
- We want to look at the “fat” sets of the covering
- Given metric space M , open covering $\mathcal{U}$, and A is sequentially compact, then $\exists \lambda > 0$
 - Suppose not.
 - Then, no $\lambda > 0$ works
 - $\lambda = 1, \lambda = \frac{1}{2}, \lambda = \frac{1}{3}, \dots, \lambda = \frac{1}{n}, \dots$ all don't work
 - $\forall \frac{1}{n} \exists p_n \in A \left(M_{\frac{1}{n}}(p_n) \right)$ is not contained in any $U \in \mathcal{U}$
 - We formed a sequence of (p_n)
 - WLOG, assume (p_n) converges to p
 - We can think of (p_n) as any convergent subsequence of another sequence in A
 - Look at point p , which is contained in U_p
 - Makes sense that there is some value λ such that $M_\lambda(p) \subset U_p$ since the points (p_n) must be inside the covering
- Assume A is sequentially compact
- $\mathcal{U}$ is an open covering of A
- $\exists \lambda$ for $\mathcal{U}$, $\mathcal{U}$ reduces to $\mathcal{U}'$ s.t. $\forall p \in A \{U_p \in \mathcal{U} : p \in A\}$
- Choose $p_1 \in A$ and $U_{p_1} = U_1$
 - If this covers A , we're done
- $p_2 \in A \setminus U_{p_1}$
- Similarly, $p_n \in A \setminus (U_{p_1} \cup U_{p_2} \cup \dots \cup U_{p_{n-1}})$
- A sequence is compact $\Rightarrow \exists p_{n_k} \rightarrow a \in A$
 - Goal here is to say p_{n_k} are some λ apart, then they can't converge
 - We know for sure $p_{n_{k+1}}$ is at least λ away from p_{n_k} as $M_\lambda(p) \subset U_{p_{n_k}}$ and $p_{n_{k+1}} \notin U_{p_{n_k}}$
 - We want to show there can be disjoint neighborhoods, so we choose $\frac{\lambda}{3}$

- Then, $M_{\frac{\lambda}{3}}(p_{n_k})$ and $M_{\frac{\lambda}{3}}(p_{n_{k+1}})$ are disjoint
- λ is fixed, but distance to a shrinks as we iterate through p_{n_k} , which gets to less than $\frac{\lambda}{3}$
- $\downarrow$ to convergence of p_{n_k}

Generalized Heine-Borel Theorem

- $A \subset M$ is totally bounded if $\forall \varepsilon > 0 \exists$ finitely many points $p_1, \dots, p_n \in A$ s.t. $A \subset \bigcup_{i=1}^n M_\varepsilon(p_i)$
- M complete $\Rightarrow A$ convergence iff closed and totally bounded $\Rightarrow A$ covered by $\mathcal{U} = \{M_\varepsilon(p) : p \in A\}$
 - Compactness means finite subcovering
- Sequentially compact $\leftarrow$ TB and closed
 - $(a_n) \in A$
 - Pugh said it isn't hard to prove, but he wanted to move onto Cantor Sets

Cantor Sets

- Remark: Invented by Henry John Steven Smith about a decade earlier
- $C \subset [0, 1]$
- Take the number line representing $[0, 1]$
- Take away the middle third, then take away the middle thirds from the remaining two thirds, and repeat this process
- Total of $\sum_{n=1}^{\infty} \frac{2^{n-1}}{3^n} = 1$ discarded
- $C = \bigcap_{n=1}^{\infty} C^n$
- $C^0 = [0, 1]$
- $C^1 = [0, \frac{1}{3}] \cup [\frac{2}{3}, 1]$
 - $C^1 = C_0^1 \cup C_2^1$
- $C^2 = [0, \frac{1}{9}] \cup [\frac{2}{9}, \frac{1}{3}] \cup [\frac{2}{3}, \frac{7}{9}] \cup [\frac{8}{9}, 1]$
 - $C^2 = C_{00}^2 \cup C_{02}^2 \cup C_{20}^2 \cup C_{22}^2$
- In this notation, 0 is for the left intervals, 2 is for the right
- Reminds us of base 3 numbers
 - $C = [0, 1] \setminus \{x : x \text{ has a base 3 expression with at least one } 1\}$
- C is perfect, which implies it's uncountable
 - $[0, 1]$ is compact, and C is closed and bounded so C is compact
 - Not sure about proof, but idea seems relatively intuitive
- C contains no (a, b) , as this interval is of finite length $b - a$, and C^n has intervals of small length
- Properties of C :
 - Nonempty, compact, perfect, totally disconnected
 - Totally disconnected means there are arbitrarily small clopen subsets
 - $C \cong C'$ (C' are other setups that create Cantor sets)
 - Every compact, nonempty set is the continuous image of C